

EE 301: Control Systems

Course code : EE 301

Course Name : Control Systems

L-T-P-C : 2.5-0.5-0-3

Prerequisites : IC 260-Signals and Systems

Intended for : UG

Elctive/Core: Discipline core for BTech in EE, Elective for BTech in CE, CSE and ME

Approval: 5th Senate, 12th Senate

Course Contents

- **Basic concepts:** Introduction, basic terminology, objective of subject, some basic examples, Notion of feedback; open- and closed-loop systems.
- **Mathematical Models:** Representation of physical systems and analogous systems, Lapalce transforms, block diagrams, transfer functions for different type of systems, block diagrams reduction techniques; Signal flow graphs and Mason's gain formula.
- **Control hardware and their models:** Potentiometers, synchros, LVDT, DC and AC servo motors, tachogenerators, electro-hydraulic valves, and pneumatic actuators.
- **Time-domain analysis:** Time domain performance criterion, transient response of first order, second order and higher order systems;
- **Steady state errors:** Static and dynamic error constants, system types, steady state errors for unity and non unity feedback systems, performance analysis for P, PI and PID controllers.
- **Frequency-domain analysis:** Bode and polar plots, frequency-domain specifications, correlation between transient response and frequency response.
- **Stability analysis:** Concept of stability by Routh stability criterion, Nyquist stability criterion, gain and phase margins, relative stability, constant M and N circles, Nichol's chart and its application.
- **Root-locus technique:** Nature of root-locus, rules of construction, root-locus analysis of control systems.
- **Compensation:** Types of compensation, Proportional, PI and PID controllers; Lead-lag compensators.
- **State-space concepts:** Eigen values and eigen vectors; Solution of state equations; Controllability; Observability; pole placement result, Minimal representations(if time permits).
- **Non-Linear systems:** Characteristics of non-linear systems, types of non-linearities, phase-plane analysis, limit cycles and describing functions (if time permits).

References:

1. Nagrath I. J. and Gopal M., **Control System Engineering.**
2. Kuo B. C., **Automatic Control Systems.**
3. Ogata K., **Modern Control Engineering.**



4. Gopal M., **Control Systems: Principle and Design.**
5. Prof. S. D Agashe, **NPTEL Video Lectures on Control Engg.**
6. Prof. M. Gopal, **NPTEL Lecture Notes on Control Systems.**
7. Dorf R. C. and Bishop R. H., **Modern Control Systems.**
8. Norman S. N., **Control Systems Engineering.**
9. IEEE Transactions on Automatic Control.
10. IEEE Transactions on Control Systems Technology.